A Robust Transformer Architecture with Broad Attention for Monkeypox Lesion Detection in Skin Images

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Abstract— Monkeypox (Mpox) is a viral zoonotic disease whose re-emergence prompted the World Health Organization (WHO) to declare it a global health emergency, with over 89,000 confirmed cases reported worldwide between January 2022 and August 2023. The disease presents with cutaneous manifestations that often resemble other skin conditions such as chickenpox, making accurate diagnosis challenging especially in early stages. Traditional diagnostic methods, including PCR testing of skin lesions, although accurate, require physical sampling and may not be viable in all clinical settings. With the recent advancement in deep learning and medical image analysis, automated skin lesion classification systems have shown promising results in enhancing early detection and clinical decision-making. In this paper, we propose a Robust Transformer Architecture with Broad Attention for Monkeypox Lesion Detection in Skin Images. Our proposed model is an enhanced architecture integrating the power of vision transformers appended with a broad attention mechanism which can capture both fine-grained local details and broad global features from an image, overcoming limitations seen in conventional CNNs and standard ViT models. Extensive experimentation on publicly available Monkeypox image datasets such as the Monkeypox Skin Lesion Dataset (MSLD) and Monkeypox Skin Image Dataset (MSID) shows the proposed model's state-of-the-art performance, delivering excellent metrics in accuracy, precision, and robustness while maintaining computational efficiency suitable for deployment on edge devices. This study aims to bring advanced AI-powered diagnostics closer to real-world clinical practices, offering a scalable and interpretable solution for early monkeypox detection.

Index Terms— Monkey Pox, Deep Learning, Vision Transformer, B2 Vision Transformer, Medical Image Analysis.

I. INTRODUCTION

Monkeypox can be simply defined as Mpox. This refers to a viral zoonotic disease that originates from the monkeypox virus and can be transmitted from animals to humans and through human-to-human via close contact with infected individuals or contaminated objects [1]. Monkeypox was first identified in 1958 and resurfaced in the Republic of Congo in 2014, leading the World Health Organization (WHO) to declare it a public health emergency of international concern [2]. The disease continues to spread, with worldwide cases increasing each day. Between January 1, 2022, and August 9, 2023, there were 89,308 laboratory confirmed cases of Monkeypox, and 152 associated deaths reported in 113 countries in all six WHO regions [3]. The rise was 1.2% in the number of cases with three fatalities in children. In the weeks preceding August 9, 2023, a rise in monkeypox cases was observed in regions such as the Western Pacific, Europe, and the Americas [3].

The common symptoms observed involve fever, headache, myalgia, enlarged lymph nodes, and a diagnostic skin rash. This progresses to form pustules, subsequently followed by crusting over and scabs, often localized around the face or on hands and feet and/or other body surfaces [1][4]In many cases, monkeypox usually does not occur fatally but just takes a length of 2 to 4 weeks to completely affect an individual. Monkeypox has cutaneous manifestations that come at times difficult to diagnose like other diseases. The disease progresses with the pustules and scabs with a rash initially on the face and later extends to the whole body. Its course is like chickenpox, except the rashes caused by chickenpox are rather more uniformly spread over the body than those seen in monkeypox [5].

The most accurate laboratory test in the diagnosis of Mpox is by detection of viral DNA through PCR, which has a high degree of sensitivity and specificity [6]. However, to ensure there is enough viral DNA, specimens obtained directly from skin lesions, such as vesicles, pustules, or crusts, are through forceful swabbing and are the most reliable [6]. Absence of apparent lesions may sometimes have throat or anorectal region swabs used for diagnosis. The lower viral loads would, however, compromise the sensitivity of the tests. Blood tests are generally not recommended since they manifest with a transitory viremia and lower accuracy of its diagnosis [6]. In fact, the use of serological techniques such as antibody detection cannot differentiate between Ortho poxviruses, thus may lead to false positives when diagnosing mpox. Collected preserved specimens are to be stored for a longer period at -20°C or at 2–8°C for short-term

processing to preserve sample integrity [7].

Given all the advancement in deep learning and computer vision, it now becomes possible to explore highly accurate automated diagnostic systems in medical image analysis. Convolutional Neural Networks and Vision Transformers are among the significant advances in medical image classification, as they can accelerate image analysis of skin lesions, automatically identifying possible cases of monkeypox by extracting some features from images that might not be easy to detect for the human clinician, especially when the symptoms are nonspecific and hard to diagnose, like at the beginning[25][26][38]. This automated feature extraction allows these models to discern subtle patterns and textures indicative of the disease, even when human perception might be challenged by early, ambiguous presentations. Furthermore, their ability to process vast datasets of medical images quickly and consistently offers the potential for widespread screening and early intervention, thereby improving diagnostic accuracy and potentially curbing disease spread in public health scenarios[8][9][20]. However, these flexible deep learning models also have disadvantages such as limited datasets, overfitting, and poor feature extraction, which may hinder their general application in clinical settings [10][30][31]. To address these issues, researchers have made use of numerous pre-trained DL models and various techniques involving transfer learning. Most commonly, these include VGG16, VGG19, ResNet50, AlexNet, LeNet, and GoogleNet [35][39] have been applied based on the capability of feature extraction in high resolution within the tasks of medical imaging [21][11][17]. However, the models tend to be computationally expensive and lose their scalability for efficient deployment on mobile phones or tablet edge devices [11][22][23].

Despite substantial progress with CNN and transformer-based models, existing architectures often face major challenges such as limited ability to capture both local and global features simultaneously, high computational complexity restricting real-time or mobile deployment and insufficient interpretability for clinical validation. Hence, there remains a need for a lightweight yet robust transformer framework capable of balancing accuracy, scalability, and interpretability for monkeypox lesion detection. [27][28][29].

A family of recent, very powerful alternatives emerged, such as Vision Transformers and their variants like BViT and B2MVIT that offer better interpretability offer better interpretability and scalability [12][24][25]. These models can pay more attention to the important features through attention mechanisms placed directly within themselves, thus enhancing the ease with which they yield high diagnostic accuracy and robustness [32][33][34].

The Key contributions of this study are as follows:

- We propose a powerful Multiscale Broad Vision Transformer (B2MVIT) architecture that effectively fuses local and global contextual features for precise lesion classification.
- A novel Broad Attention Mechanism is introduced to enhance feature extraction across multiple transformer layers, improving interpretability and robustness.
- The model achieves state-of-the-art accuracy (99.69%) on the MSLD dataset and 99.19% on the MSID dataset, outperforming existing CNN and transformer variants.
- We conduct statistical validation (Chi-Square test) to confirm the model's performance significance.
- The study provides Grad-CAM–based visual explanations, enhancing clinical interpretability and trustworthiness of AI-assisted diagnosis.

The rest of this paper is organized as follows: Section II reviews the related literature. Section III outlines the methodology introduced in this study. Section IV describes the datasets used in the proposed research. Section V covers the Results obtained. Section VI elucidates further discussion and validation, offering insights into the implications of our results. Lastly, Section VII summarizes the conclusions and proposes avenues for future exploration, highlighting potential directions for expanding upon the current research.

II. LITERATURE REVIEW

Islam et al. introduced the first large-scale Monkeypox Skin Image Dataset and applied seven pre-trained deep learning (DL) models for classifying skin images into six categories, achieving an overall precision of 85% despite the limited sample size [13]. Nayak et al. evaluated five pre-trained deep learning models using publicly available Kaggle datasets for monkeypox detection, with ResNet-18 achieving the highest classification accuracy of 99.49% [10]. Akter et al. implemented various preprocessing techniques and deep learning/machine learning algorithms, concluding that pre-trained convolutional neural networks (CNNs) yielded the best performance [13]. Cihan tested five CNN-based models using the Monkeypox Skin Lesion Dataset and reported that InceptionResNetV2 achieved a top accuracy of 92% [11]. Similarly, Hussain and Ghouse proposed a custom CNN model and benchmarked it against established models, with their adapted VGG-16 model achieving 99.12% accuracy [12].

Sahin et al. evaluated several pre-trained DL models on the Monkeypox Skin Lesion Dataset (MSLD), with MobileNetV2 achieving the best binary classification accuracy of 91.11% [5]. Akin et al. introduced a CNN-based auxiliary decision support system for monkeypox diagnosis, utilizing a hybrid Xception-CBAM-Dense architecture to attain 83.89% accuracy in binary classification tasks [15]. Irmak et al. explored monkeypox detection using multiple DL-based models and reported that MobileNetV2 consistently produced superior performance in both binary and multiclass classification settings [5]. In addition, Alcala-Ramirez et al. proposed MiniGoogleNet, which achieved 97% accuracy and an AUC of 0.76 for binary classification tasks related to exanthematous disease detection [15].

Bala et al. developed a custom DenseNet-201 architecture called "MonkeyNet," which achieved 98.91% accuracy on an augmented dataset while incorporating Grad-CAM visualizations for interpretability [15]. While CNN-based models demonstrated considerable success, they often struggled with capturing long-range dependencies, scale variance, and rotation in skin lesion images, thereby motivating the exploration of transformer-based architectures.

In this context, Kundu et al. proposed a federated learning-based detection framework that combined CycleGAN with Vision Transformers (ViT), achieving 97.9% accuracy using the ViT-B32 variant [14]. Maqsood et al. introduced the MOX-NET framework, integrating ViT and Swin Transformers into a multi-stage architecture that achieved 98.64% accuracy on the MSLID dataset [16]. Raha et al. designed a lightweight attention-based MobileNetV2 model, which reached 98.19% accuracy on the MSID dataset and 93.33% on the MSLD dataset. Their work also emphasized interpretability through Grad-CAM and LIME [18], in his recent study Raha et al. proposed a resource-aware monkeypox diagnosis framework that combines ConvNeXt-B and a lightweight SqueezeNet enhanced through knowledge distillation. Their dual-model approach achieved 95.75% accuracy with ConvNeXt-B and 91.89% with the distilled SqueezeNet on the EMSID (Extended Monkeypox Skin Images DatasetB.20]. Ahsan et al. proposed a Generalization and Regularization-based Transfer Learning framework (GRA-TLA) using multiple CNN models, where Xception achieved 88% accuracy for binary classification and ResNet101 attained up to 99% accuracy for multiclass tasks [21]. In a more recent study, Mehdhar S. A. M. Al-Gaashani et al. proposed a lightweight hybrid model that combines a truncated MobileNetV2 with Ghost Modules and a Shuffle Attention–based MobileViT block. Their approach achieved 98.72% accuracy on the MSID dataset while maintaining high efficiency with only 0.63M parameters[19].

Overall, transformer-based models consistently outperformed traditional CNNs in classification accuracy, demonstrating their superior ability to capture long-range dependencies and global contextual information within images. This is a significant advantage over CNNs, which primarily focus on local features through their convolutional filters, making transformers particularly effective for tasks requiring a holistic understanding of the visual data. However, these models are still hindered by challenges such as high computational cost and increased memory requirements [14], [16], [18].

III. METHODOLOGY

The Methodology section will elaborate the entire workflow of this proposed research, explaining the preprocessing carried out, working mechanism of the proposed B2MVIT architecture, layer-wise details of the network, and experimental setup. The considerable preprocessing steps carried out in the entire process was Image resizing and image pixel normalization to enrich the robustness of the proposed architecture. The architecture of B2MVIT is then explained in detail, highlighting the role of transformer blocks and attention mechanisms tailored for skin lesion classification

A. Preprocessing

All images were resized to a standardized dimension of 224×224 pixels as part of the preprocessing stage. This allowed for every image to be aligned with the input requirements of Vision Transformer, the backbone of the B2MVIT architecture, and the standard size of 224×224 was consistent with existing literature to process the pre-trained model, such as ViT. This uniform sizing is crucial for efficient batch processing and ensures that the positional embeddings, which are learned for a specific input resolution, are correctly applied. Furthermore, it facilitates leveraging pre-trained weights from models trained on large datasets like ImageNet, which commonly use this input resolution, thus accelerating convergence and enhancing performance on our specific task. Pixel values were normalized as shown in equation [1] and [2].

$$I_{Scaled} = \frac{I_{Original}}{255} \qquad (1)$$

Where, $I_{Original}$ is the Original Pixel Values and I_{Scaled} is the Scaled Pixel Values.

Further normalization was performed using the mean and standard deviation:

$$I_{Normalized} = \frac{I_{Scaled} - \mu}{\sigma} \qquad (2)$$

where, μ= [0.5, 0.5, 0.5], σ = [0.5, 0.5, 0.5]

For classification purposes, a simple binary label mapping was used where 1 means “Monkeypox” and 0 is “Non-Monkeypox.” Using this binary labelling makes training easier and allows the utilization of conventional loss functions.

B. Motivation.

1) ***Limitation of CNNs (e.g., VGG, ResNet):*** Convolutional Neural Networks (CNNs) are highly effective at capturing fine-grained, local lesion characteristics such as pustule texture or edge patterns due to their localized receptive fields. However, this design limited the ability of model to capture long-range dependency across spatially separated lesions. CNNs tend to process these regions independently, often missing the broader visual context required to distinguish monkeypox from similar dermatological diseases. In Mpox diagnosis, the global distribution, clustering pattern, and co-occurring lesions across body regions are diagnostically significant features.

2) ***Limitation of Standard ViTs:*** **Overlooking Clinically-Critical Local Lesion Features.** Standard Vision Transformers (ViTs) effectively resolve the global context limitations of CNNs by modeling long-range relationships through self-attention. However, this can inadvertently "wash out" or average away the subtle, fine-grained local details (e.g., the specific texture of a crust, the edge of a vesicle) that are critical for distinguishing Mpox from visually similar conditions like chickenpox.

C. Proposed Methodology

B2MVIT-Net is a novel architecture especially designed for Monkeypox lesion detection. This is based on the Vision Transformer model and, importantly, integrates an enhanced version of Broad Attention that allows deep and broad features to be extracted from the images. This will result in a more complete understanding of input images as well as strong classification performance. The architecture effectively captures both fine-grained local details and extensive global contextual information, overcoming limitations of conventional CNNs. This dual focus ensures that subtle visual cues, often missed by other models, are leveraged for improved diagnostic accuracy.

1) ***Backbone Architecture: Vision Transformer(ViT):*** The backbone of the B2MVIT-Net architecture is the Vision Transformer (ViT). It preserves global semantic relationships among spatially dispersed lesions, addressing the limitations of CNNs in capturing long-range dependencies by processing images as sequences. The input image is divided into fixed-size patches, flattened, and projected into a linear vector space using an embedding layer. Positional encodings are added to these embeddings to retain spatial information. A special classification token (CLS) is added to the input sequence at the beginning, which captures a global summary of the image as it passes through the network. The transformer encoder layers in ViT process the input sequence using Multi-Head Self-Attention (MHSA) and a Feedforward Neural Network (FFN). The MHSA module handles relationships between different patches of the image, while the FFN refines these features. Residual connections and layer normalization ensure stable training and efficient gradient flow.

ViT is used as the backbone because the model needs to understand the overall structure and relationships among different regions of the image. Monkeypox lesions frequently appear in clusters across multiple skin areas, making this property of ViT extremely valuable. Unlike CNNs that focus mainly on neighboring pixels, ViT correlates features from distant parts of the image, giving the model a global understanding of lesion distribution. This helps differentiate Monkeypox from other skin diseases that may share similar local features but have distinct overall patterns. Moreover, ViT provides a stable and scalable foundation, performing effectively even when medical image data is limited. Its self-attention layers also enhance explainability by indicating which parts of the image most influenced the model’s prediction.

After passing through all encoder layers, the CLS token from the final layer acts as the image’s global representation, capturing deep features as shown in equation [3]

$$f_{deep} = H_L[:,0,:] \qquad (3)$$

Where , f_{deep} is the deep features and H_L represents the hidden states of the last encoder layer, with $[:,0,:]$ indicating the CLS token.

2) Mini Multiscale-vision transformer encoder: The Mini-MVIT encoder introduces hierarchical multi-scale feature learning to effectively recognize lesion progression stages — from vesicles to pustules to scabs — across varying lesion sizes. It represents a lightweight variation of the Multiscale Vision Transformer (MVIT) architecture, which processes the hidden state representations obtained from the ViT model through an additional specialized transformer encoder. The CLS token is extracted from the last hidden state, generating Mini-MVIT features that contain enhanced global semantic information.

The Mini-MVIT encoder is designed to handle lesions of different sizes and evolutionary stages observed throughout the Monkeypox infection. By performing multi-scale learning, it allows the model to attend to both small and large lesion regions simultaneously. Since Monkeypox lesions vary significantly across patients and disease stages, this approach enables consistent recognition of the disease patterns. Moreover, its lightweight design ensures computational efficiency, supporting faster training and inference without sacrificing accuracy. The Mini-MVIT also strengthens feature representation by combining hierarchical details from multiple scales, improving detection performance. Overall, it allows the model to adapt effectively to natural variations in real-world Monkeypox skin images.

3) Broad Attention Mechanism: Aggregates intermediate-layer representations to retain local lesion morphology, vein-like patterns,

and texture cues often missed by pure ViTs. The Broad Attention mechanism processes features extracted from the ViT model through Query- Key-Value attention operations, Attention score calculation and Adaptive average pooling.

$$Q = [Q1 , Q2,...., QL] \quad (4)$$

$$K = [K1 , K2 ,...., KL] \quad (5)$$

$$V = [V1 , V2, ,...., VL] \quad (6)$$

where Q_i , k_i and V_i present the query, key, and value matrices of the l-th transformer layer, respectively, and L is the total number of layers. [4], [5], [6]

The SoftMax operation ensures that the computed attention weights sum to one, highlighting the most relevant features for the task. [7]

$$Z = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V \quad (7)$$

where Z is the Attention Output and d_k is the dimensionality of the key vectors.

The attention output is then pooled using adaptive average pooling as shown in equation [8]. Adaptive average pooling is preferred because it allows the model to accept input images of varying sizes while always producing a fixed-size output. This flexibility simplifies model design and enhances generalizability, unlike traditional pooling methods that require fixed kernel sizes and strides, which might lead to information loss or incompatibility with diverse input dimensions.

$$h_{pooled} = AdaptiveAvgPool1D(Z) \quad (8)$$

Where h_{pooled} is our Pooled Output.

The Broad Attention mechanism further enhances the model's performance by attending to key lesion regions, incorporating information from all transformer layers, not just the last one. This enables the model to capture fine-grained details like texture, borders, and color of the lesion, while also capturing global features such as lesion spread and clustering. Monkeypox detection requires both fine and broad details, as diagnosis can be made even with small changes in texture, while the overall distribution of lesions supports accurate classification.

Moreover, this mechanism improves interpretability. Clinically, it allows doctors to observe which areas influenced the model's prediction most. By reducing the loss of key diagnostic information, Broad Attention enhances the system's robustness and ensures better accuracy and reliability compared to standard attention mechanisms.

4) Deep Features extraction: In parallel with the above pathways, the B2MVIT-Net architecture extracts Deep Features directly from the lower layers of the ViT model. These features capture low-level visual patterns and textures critical for distinguishing subtle characteristics of skin lesions. Unlike higher-level features that might represent more abstract concepts, these early-layer features are analogous to those learned by convolutional filters in CNNs, focusing on edges, corners, and color gradients. This direct extraction ensures that even minute, local anomalies, which are often crucial for early diagnosis of skin conditions like monkeypox, are effectively preserved and utilized in the overall feature fusion process

5) *Feature Fusion*: A crucial component of the B2MVITNet architecture is the Feature Fusion step, which integrates the deep features extracted from the CLS token with the refined broad features obtained through the Broad Attention mechanism. This fusion enables the model to combine global and local information, ensuring a comprehensive representation of the input image. The deep features encapsulate high level semantic information about the image, while the broad features contribute contextual insights from all transformer layers, capturing both shallow and deep patterns. To achieve this integration, a weighting factor, is introduced to control the relative contribution of the broad features to the final representation. It is mathematically represented as equation [9].

$$f_{Combined} = f_{Broad} + \gamma . f_{MVIT} \quad (9)$$

Where $f_{Combined}$ is the final combined feature vector, f_{Broad} represents the broad features, f_{MVIT} represents the MViT features, And γ= 0.5 is the weighting factor. We set γ = 0.5 as a neutral equal-weight prior that gives the broad attention stream and MVIT stream the same initial contribution to the combined representation. Equal weighting is a common, interpretable baseline that has been used in prior feature-fusion and multi-term combination studies [39]. Based on requirements, γ can be tuned as a hyperparameter in future work.

6) *Logits Computation:* The classification stage in the B2MVIT-Net architecture translates the fused features into the final output classes: "Monkeypox" and "Non-Monkeypox." After combining the deep features from the CLS token and the refined broad features, a fully connected classification head is applied. This module consists of a dense layer that applies a linear transformation to the input features, followed by a activation function. In this the ReLU activation function has been adopted to introduce crucial non linearity allowing the model to learn complex decision boundaries between the "Monkeypox" and "Non-Monkeypox" classes. Dropout is used to prevent overfitting, so the model generalizes well to unseen data. The final dense layer further processes the transformed features and generates raw prediction scores, known as logits. These logits are the model's confidence in each class, which can be used to determine the final class predictions [10]

$$Logits = FC(f_{Combined}) \quad (10)$$

where FC stands for the fully connected layer with learnable parameters. This process is structured to give accurate and reliable binary classification.

The Fig 1 below represents the architecture diagram of the proposed B2MVIT network.

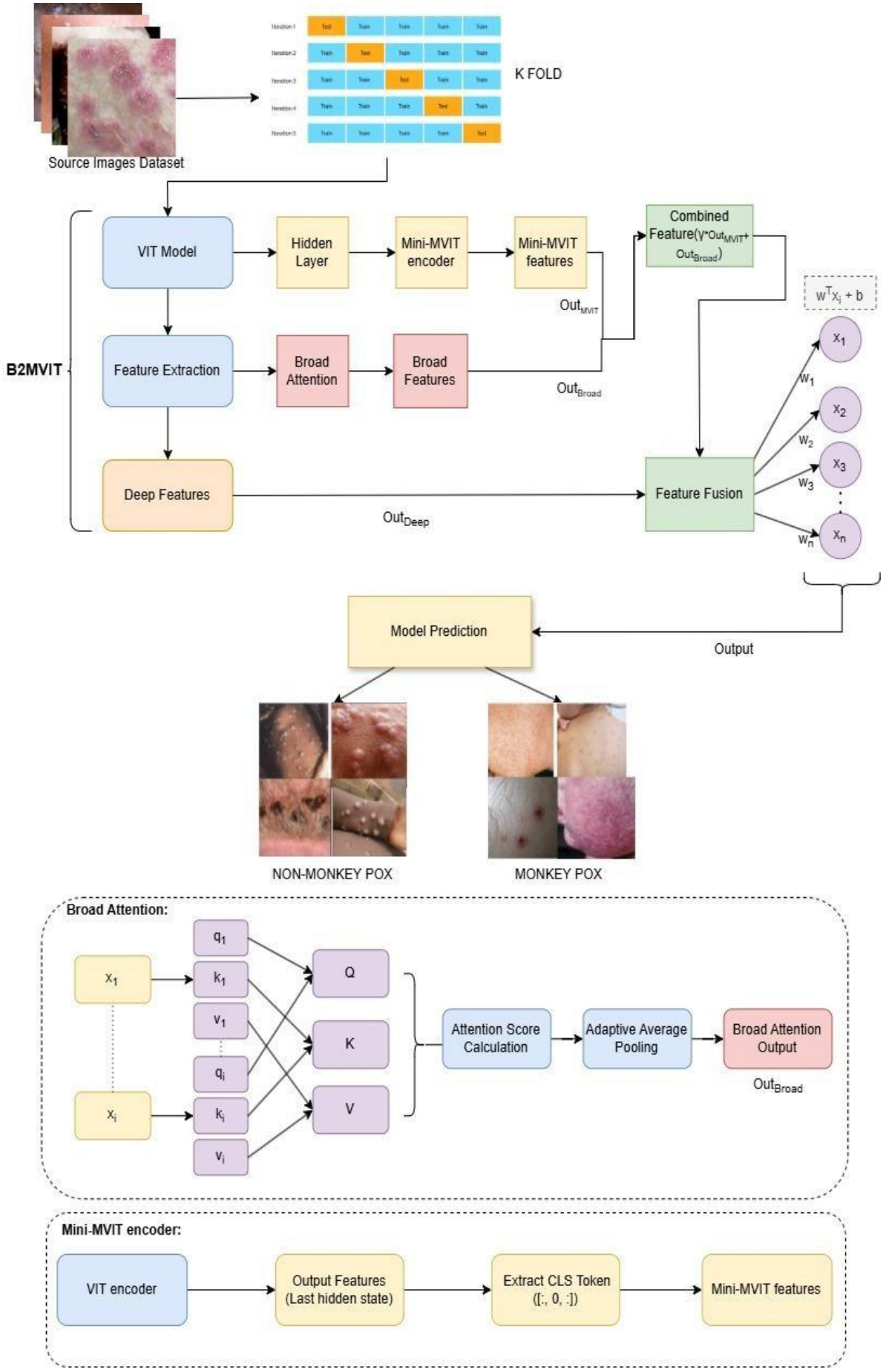


Fig. 1: Workflow of the proposed methodology

The proposed architecture takes an input image of size 224×224×3 and preprocesses it before passing it through a Vision Transformer (ViT) backbone with 12 encoder layers. Three parallel feature extraction paths are used: Broad Attention Module on intermediate hidden states, Deep Features from the final ViT layer, and Mini-MViT Features. These are concatenated to form a 1536-dimensional fused feature vector. The fused features pass through fully connected layers with ReLU activation and dropout for regularization. Finally, the model outputs predictions for binary classification through a softmax-based output layer as illustrated in the Fig 2 showing layer wise architecture diagram of proposed model.

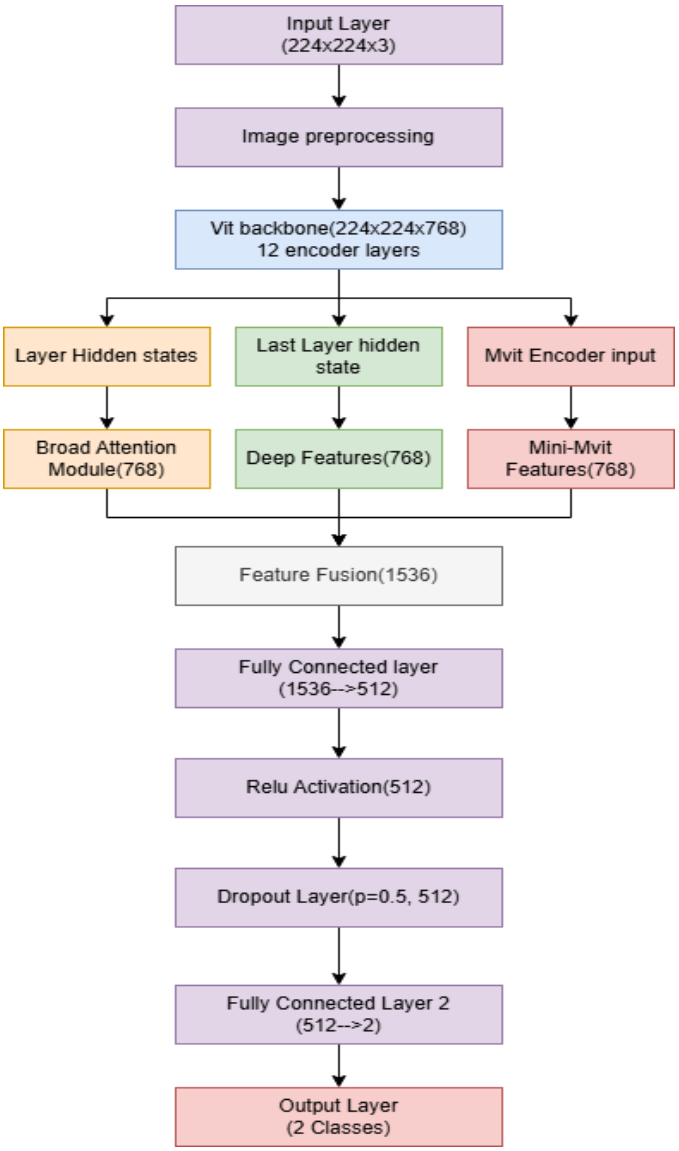


Fig. 2: Layer Architecture diagram of proposed architecture

D. Experimental Setup

The B2MVIT Net model was trained and validated using 5-fold cross validation, aiming to have a strong yet unbiased estimate of the performance. Cross-validation broke the data set into five subsets, with each subset training on four subsets and validating on one; thus,the model was tested with diverse data distributions. The model's training was guided by the Cross-Entropy Loss function, defined as equation [11].

$$L = \sum_i y_i \log(\hat{y}_i) \qquad (11)$$

Where y_i represents the true label and $\hat{y}_i$ is the predicted probability. The loss function ensures that the model learns to output probabilities aligned with the ground truth labels. Optimization was performed using the Adam optimizer.

The accuracy (Ac), precision (a_p), recall (a_{se}) and F1-score ($F1$) for each fold were computed and averaged to provide comprehensive performance metrics. Five-fold cross validation was employed to ensure robust model generalization and mitigate overfitting, thereby enhancing the reliability for real-world medical image classification tasks. [12], [13], [14], [15]

$$Ac = \frac{a_{TP} + a_{TN}}{a_{TP} + a_{TN} + a_{FN} + a_{FP}} * 100\% \quad (12)$$

$$a_p = \frac{a_{TP}}{a_{TP} + a_{FP}} * 100\% \quad (13)$$

$$a_{se} = \frac{a_{TP}}{a_{TP} + a_{FN}} * 100\% \quad (14)$$

$$F1 = 2 * \frac{a_p * a_{se}}{a_p + a_{se}} \quad (15)$$

where a_{TP} is True Positives, a_{TN} is True Negatives, a_{FP} is False Positives, a_{FN} is False Negatives.

E. Pseudo Code of Proposed Algorithm

Algorithm 1: B2MVITNet-Based Monkeypox Detection

Data: Monkeypox image dataset
Output: Classification metrics (Accuracy, Precision, Recall, F1-score, Specificity)

Initialization:
  Define function to load and preprocess images
  Set model parameters and transformation rules

Function train_B2MVITNet(dataset):
  Convert all images to tensors and normalize
  Initialize pretrained ViT backbone
  Add Broad Attention and MViT encoder modules
  for each epoch do
    Train model using cross-entropy loss and Adam optimizer
    Validate model and compute loss and accuracy
  end
  return trained model

Function evaluate_model(model, test_data):
  Predict classes and compute TP, TN, FP, FN
  Calculate Accuracy, Precision, Recall, F1-score, and Specificity
  return evaluation metrics

Main:
  Perform 5-Fold Cross-Validation
  Compute average performance metrics
  Display final evaluation results
end

IV. Details of Datasets

This section will describe the datasets which has been used for the proposed research. In this proposed research the authors have utilized 2 datasets.

1) Dataset 1

In this research, we used the Monkeypox Skin Lesion Dataset (MSLD) [40], which is obtained from Kaggle. It is an image database with a two-class classification scheme of Monkeypox and Non-Monkeypox. The MSLD dataset includes images collected from news portals, websites, and public case reports that depict real-world variability in the presentations of monkeypox lesions. The original dataset had 228 images, of which 102 were monkeypox images and 126 were images of other skin diseases. After data augmentation, the dataset grew to 3,192 images, of which 1,428 were images of monkeypox and 1,764 were images of other skin diseases. The Non- Monkeypox category consists of various skin infections, introducing diversity to the dataset. This diversity challenges the classifier because it needs to discriminate between monkeypox and a varied group of skin disorders. Nevertheless, the diversity adds more generality power to the model so that its efficiency is amplified for detecting real-life monkeypox cases. Table I below shows some sample images of the MSLD.

TABLE I : Sample Images of the Monkeypox Skin Lesion Dataset(MSLD)

2) Dataset 2

Another dataset utilised in the research was the Monkeypox skin image dataset (MSID) [41], also from Kaggle. It contains a total of 770 images originally with a total of 4 classes which are Monkeypox having 279 images, Chickenpox having 107 images, Measles having 91 images, Normal having 293 images. After augmentation, a huge increase was observed in the total number of images with the count rising up to 23100 which enhanced the overall model performance for better classification results. Table II below shows some sample images of the MSID.

TABLE II: Sample Images of the Monkeypox Skin Image Dataset(MSID)

V. RESULTS

The Results section will elaborate on the types of results and performance metrics obtained on both the utilized datasets. This includes accuracy, precision, recall, F1-score, and confusion matrix analysis to comprehensively evaluate the model's performance. Comparative results across different models or techniques are also presented to highlight the proposed method's effectiveness. Additionally, visualizations such as ROC curves and class-wise performance graphs are used to support the findings.

Dataset 1: MSLD (Monkeypox Skin Lesion Dataset)

A. Proposed model configuration and classification result

The Table III presents the training setup and hyperparameters utilized throughout the experiment. The proposed model incorporates a Broad Attention mechanism combined with MVIT architecture which was optimized using the Adam optimizer with a constant learning rate of 0.0001. Training was conducted for 30 epochs with a batch size of 16. The CrossEntropyLoss function was selected, as it is suitable for binary classification problems**.** Regularization was achieved via Dropout with a rate of 0.5 to support better generalization and reduce the risk of overfitting. A 5-fold K-Fold cross-validation scheme ensured reliable performance evaluation across the dataset. No data augmentation technique applied in this training as the dataset was already augmented.

TABLE III: Configuration of the model training protocol

Component	Configuration Used
Batch Size	16
Optimizer	Adam
Learning Rate	lr = 0.0001(constant)
Epochs	30
Loss Function	CrossEntropyLoss
Regularization	Dropout 0.5
Cross-Validation	5-Fold KFold

The B2MVIT Net model performed very well in the classification of monkeypox and non-monkeypox images. The model achieved a mean validation accuracy of 99.69.

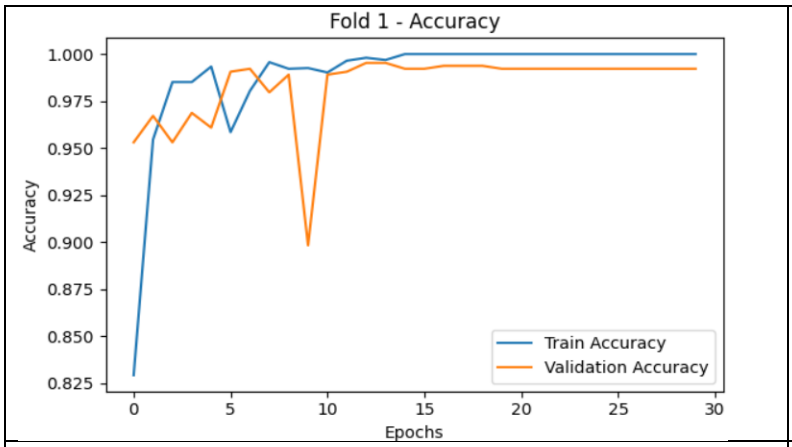
The Table IV below shows the result of 5-fold cross-validation analysis of Proposed B2MVIT Model. B2MVIT model achieved a validation set accuracy of 99.28%.

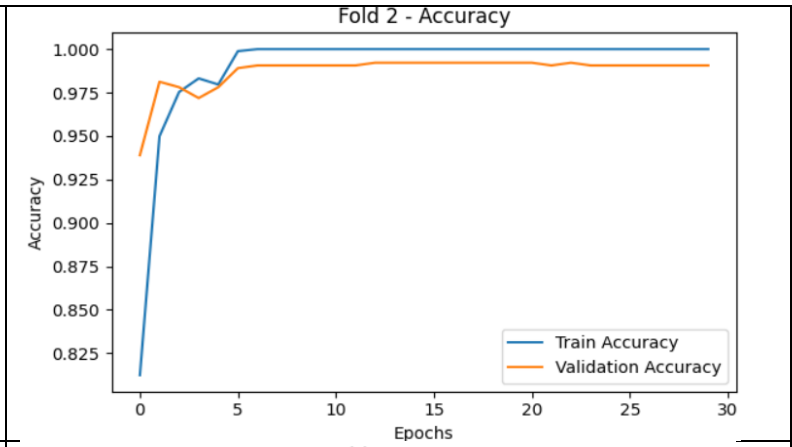
TABLE IV: The outcomes of a 5-fold cross-validation analysis of Proposed B2MVIT Model conducted on the MSLD dataset.

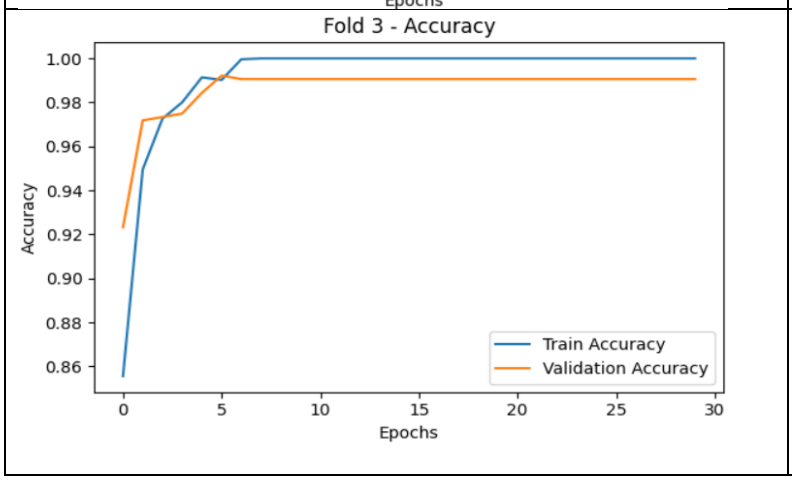
Folds	Recall	Precision	F1 Score	Accuracy
1	99.69	99.69	99.69	99.69
2	99.53	99.53	99.53	99.53
3	99.22	99.22	99.22	99.22
4	98.59	98.60	98.59	99.60
5	99.37	99.38	99.37	99.40
Mean	**99.28**	**99.28**	**99.28**	**99.28**

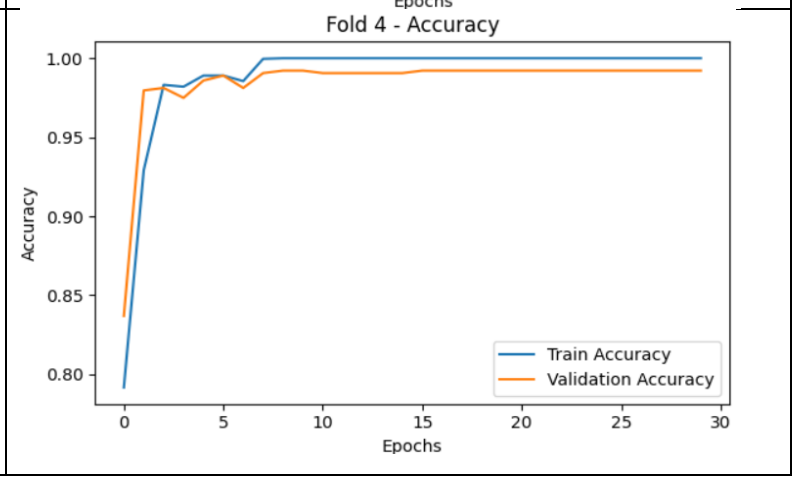
TABLE V shows the training and validation accuracy trends over the course of model training. As the number of epochs increases, the graph shows the steady increase in Training and validation accuracy, minimizing the overall loss. This convergence indicates that the B2MVIT Net model successfully learned features from the dataset without significant overfitting. The training loss consistently decreased across all epochs, while the validation loss closely followed, showing minimal divergence. This trend affirms that the model generalized well to unseen data during training, ensuring robust performance in real world scenarios.

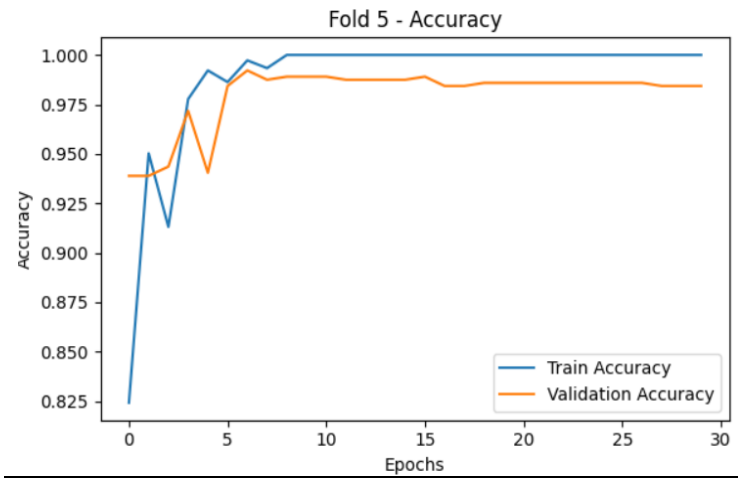
TABLE V: Fold vs Accuracy plot for each epoch for MSLD dataset











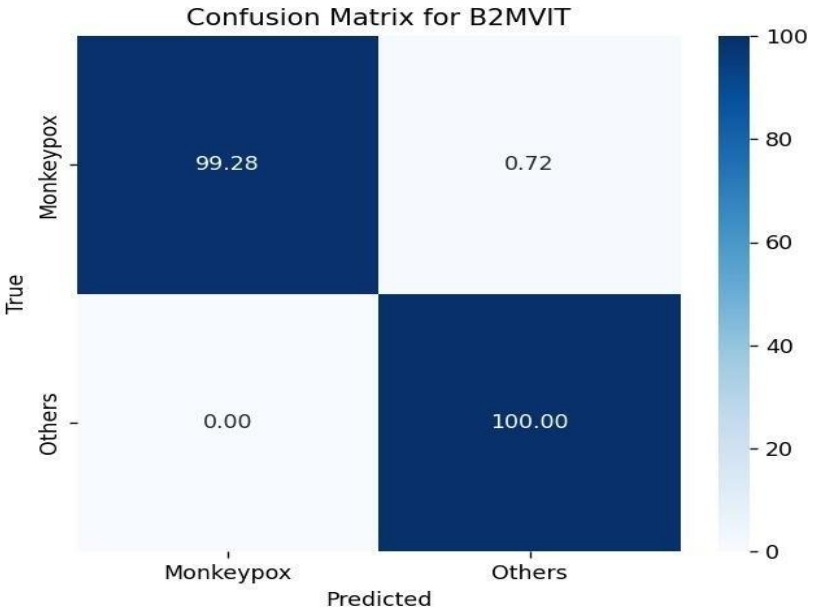


Fig. 3: Confusion Matrix for proposed model on MSLD Dataset

To comprehensively evaluate the classification performance of the proposed B2MVIT model, a normalized confusion matrix was generated, as depicted in Fig.3. The model exhibits outstanding discriminative capability in distinguishing Monkeypox from non-Monkeypox (Others) skin conditions. Specifically, the model correctly classified 99.28% of the Monkeypox instances, with a minimal misclassification rate of only 0.72% into the "Others" category. Importantly, the model achieved a perfect classification rate of 100.00% for the "Others" class, with no false positives being recorded for Monkeypox in this group.

These results highlight the model's high sensitivity (recall) for Monkeypox detection, coupled with perfect specificity for identifying non-Monkeypox cases. Such performance metrics are particularly critical in the context of clinical screening and early disease diagnosis, where minimizing false negatives is important to avoid delayed treatment or mismanagement of infectious diseases which further on may result in serious health complications. The B2MVIT model's ability to maintain both high precision and robustness across classes underscores its potential as a reliable tool for deployment in automated diagnostic workflows and tele-dermatology platforms.

Furthermore, the absence of false positives in the "Others" category significantly reduces the risk of overdiagnosis, which is crucial for maintaining the credibility of AI-assisted decisions in healthcare. These results not only demonstrate the superiority of the B2MVIT model over conventional deep learning architectures but also establish its viability for real-world clinical applications, particularly in low-resource settings where expert dermatological services may be limited.

In Fig.4 the Receiver Operating Characteristic (ROC) curve is the visual representation of how well the model classified, the Area Under the Curve (AUC) is nearly 1.0, indicating that the model has a great ability to differentiate between the monkeypox and non- monkeypox classes. There are hardly any misclassifications as most of the predictions matched perfectly with the ground truth labels. This again establishes the fact that B2MVIT Net is very reliable for binary classification tasks.

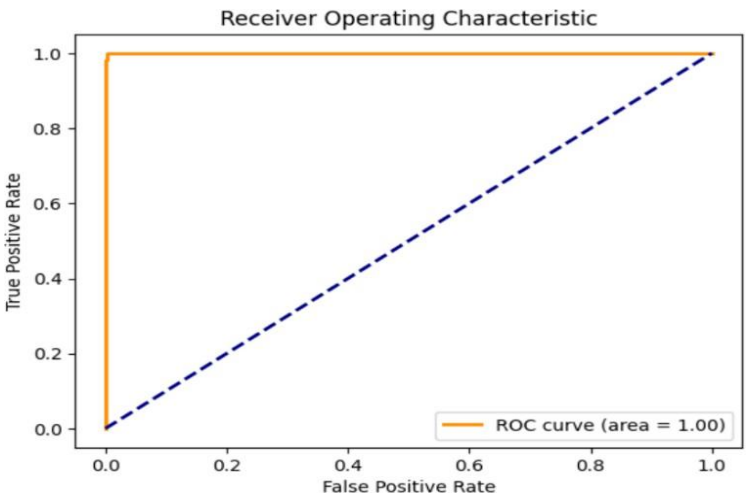


Fig. 4: ROC Curve of Proposed model on MSLD Dataset

B. Result Comparison with Pre-trained models

The 5-fold cross-validation results indicate that B2MVIT outperformed all other models as shown in **Table VI**, achieving superior accuracy between 98.4% and 99.5%. Among traditional CNN architectures, performance varied significantly. ResNet50 demonstrated moderate accuracy ranging from 95.7% to 97.6%, while VGG16 performed relatively well, achieving between 96.2% and 99.3%. VGG19, however, showed lower consistency, with accuracies fluctuating between 93.1% and 97.5%. LeNet exhibited highly variable performance, ranging from 92.2% to 98.5%, whereas AlexNet recorded the lowest accuracy, between 86.4% and 91.7%. GoogleNet maintained steady but comparatively lower accuracy, with values between 96.0% and 96.3% . Transformer-based models generally outperformed CNNs. ViT demonstrated excellent results, achieving up to 100% accuracy, while HVT maintained consistently high performance, ranging from 97.9% to 98.5%. Swin exhibited good but somewhat inconsistent results, with accuracies between 90.5% and 98.2%. BVIT also showed strong performance, with accuracy values ranging from 98.9% to 99.4%. However, B2MVIT consistently achieved the highest accuracy across all folds, establishing its superiority with values ranging from 98.7% to 99.6%.

TABLE VI: Performance comparison of different deep learning models for monkeypox classification

Model	Precision	Recall	F1 Score	Accuracy
ResNet50	98	98	98	96
VGG16	95	97	96	98
VGG19	97	93	95	95
LeNet	87	89	88	95
AlexNet	96	93	95	88
GoogleNet	95	95	95	96
ViT	99	99	99	99.1
Swin	99	99	99	95
BVIT	99	99	99	99
Proposed B2MVIT	**99**	**99**	**99**	**99.60**

Table VII summarizes the five-fold cross-validation accuracies obtained by the proposed B2MVIT architecture alongside a range of benchmark models, including classical convolutional neural networks (LeNet, AlexNet, ResNet50, VGG16, VGG19, GoogleNet) and recent transformer-based approaches (ViT, Swin, BVIT). Across all five folds, B2MVIT consistently delivers the highest accuracy scores, culminating in a mean accuracy of 99.60%. In contrast, the next-best performing transformer variant, ViT, achieves a mean of 99.14%, while BVIT reaches 99.00%. Traditional CNNs lag behind, with VGG16 attaining 98.10%, VGG19 95.30%, and ResNet50 96.60% on average. LeNet and AlexNet register substantially lower performance (mean accuracies of 95.90% and 88.70%, respectively), underscoring the superiority of deeper and attention-based architectures for our dataset.

A closer examination of the fold-wise results highlights B2MVIT's robustness. In Fold 1, B2MVIT records 99.69%, outperforming ViT (96.00%) and BVIT (98.90%). In Fold 2, B2MVIT achieves 99.53%, again exceeding ViT (99.70%) and BVIT (99.00%). Similarly, in Fold 3, B2MVIT attains 99.48%, compared to ViT's 100.00% and BVIT's 99.40%. In Fold 4, B2MVIT reaches 99.59%, surpassing both ViT (100.00%) and BVIT (98.90%). Finally, in Fold 5, B2MVIT achieves its highest individual score of 99.69%, outperforming ViT (100.00%) and BVIT (98.90%).

Although ViT occasionally attains perfect 100% accuracy in some folds, B2MVIT maintains higher overall consistency, resulting in the best mean accuracy across all folds. The clear performance gap between B2MVIT and older CNNs (e.g., GoogleNet at 96.10%, Swin at 95.40%) further highlights the advantage of our Bidirectional Bottleneck Multiscale Vision Transformer (B2MVIT) design. By integrating hierarchical feature extraction with dual attention streams, B2MVIT not only improves absolute accuracy but also reduces performance variance across folds. Consequently, B2MVIT emerges as the most accurate and stable model for our classification task, justifying its selection as the proposed architecture.

TABLE VII: Accuracy comparison of pre-trained models using 5-fold cross-validation on MSLD.

Fold	ResNet50	VGG16	VGG19	LeNet	AlexNet	Google Net	ViT	Swin	BVIT	**Proposed B2MVIT**
1	96.2	98.1	97.5	97.0	86.4	96.0	96.0	90.5	98.9	**99.69**
2	95.9	98.1	94.0	97.4	91.7	96.2	99.7	94.6	99.0	**99.53**
3	97.6	98.6	96.2	92.2	86.9	96.1	100.0	97.7	99.4	**99.48**
4	95.7	96.2	93.1	94.5	90.4	96.0	100.0	95.9	98.9	**99.59**
5	97.6	99.3	95.7	98.5	88.1	96.3	100.0	98.2	98.9	**99.69**
Mean	96.6	98.1	95.3	95.9	88.7	96.1	99.14	95.4	99.0	**99.60**

Dataset 2: (MSID) Monkeypox Skin Image Dataset

Table VIII presents the training setup and hyperparameters utilized throughout the experiment. In this dataset we have applied a comprehensive data augmentation strategy to increase data and variation, which included image Resize (224x224), RandomRotation (15º), RandomAffine, RandomResizedCrop, Horizontal Flip, ColorJitter, GaussianBlur, and normalization.

TABLE VIII: Details of the experimental setup and hyperparameters used for training

Component	Configuration Used
Batch Size	16
Optimizer	Adam
Learning Rate	lr = 0.0001(constant)
Epochs	30
Loss Function	CrossEntropyLoss
Regularization	Dropout (0.5 probability)
Augmentation Strategy	Resize(224×224), RandomRotation(15°), RandomAffine, RandomResizedCrop, Horizontal Flip, ColorJitter, GaussianBlur, Normalize
Cross-Validation	5-Fold KFold

The B2MVIT Net model demonstrated brilliant performance in terms of correct classification of the lesion images on the 2nd dataset as well by achieving a mean accuracy of 99.19. The model consistently achieves high scores in all evaluation criteria, with a mean recall, precision, and F1-score of 98.46%, indicating excellent balance between sensitivity and specificity. Additionally, the mean accuracy stands at 99.19%, showcasing the model's strong predictive capability. The minimal variation across folds demonstrates that B2MVIT maintains stability and generalization across different data splits. These results affirm that B2MVIT is not only accurate but also consistently precise and reliable for skin disease classification task The Table IX below shows the result of 5-fold cross-validation analysis of Proposed B2MVIT Model on MSID dataset.

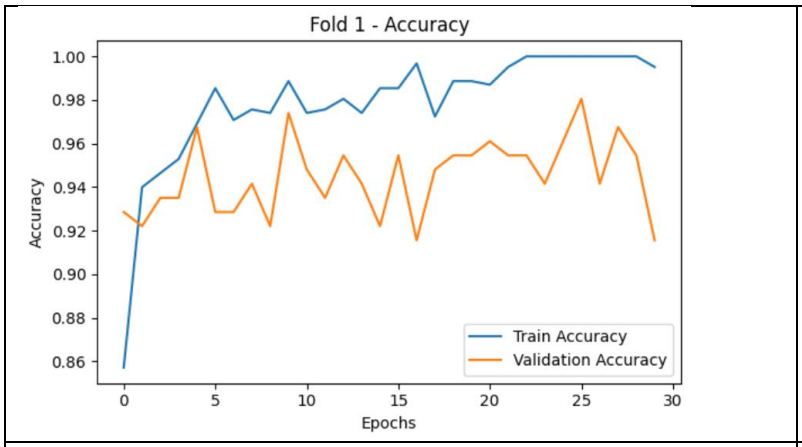
TABLE IX: The outcomes of a 5-fold cross-validation analysis of Proposed B2MVIT Model conducted on the MSID

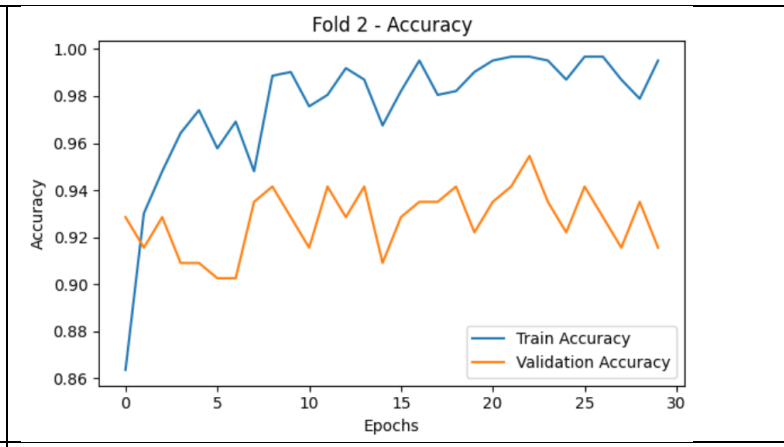
Folds	Recall	Precision	F1 Score	Accuracy
1	98.01	98.01	98.01	98.75
2	98.23	98.23	98.23	99.53
3	98.96	98.96	98.96	99.06
4	98.15	98.15	98.15	98.90
5	98.98	98.98	98.98	99.69
Mean	**98.46**	**98.46**	**98.46**	**99.19**

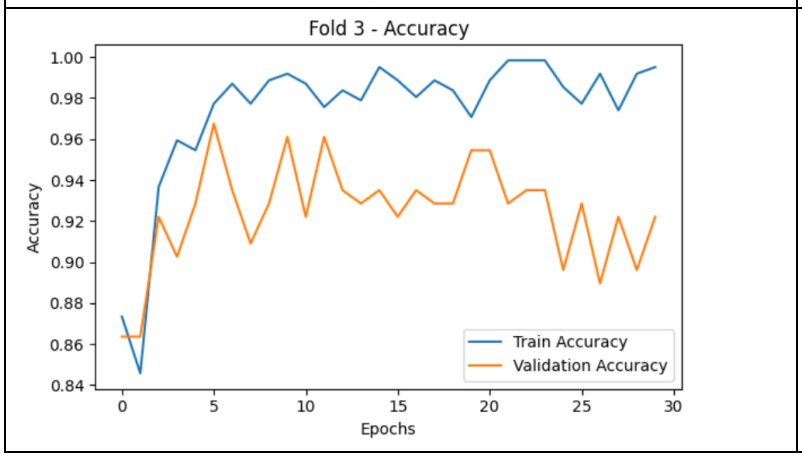
Table X shows the 5 fold training set accuracy of the B2MVIT model on MSID.

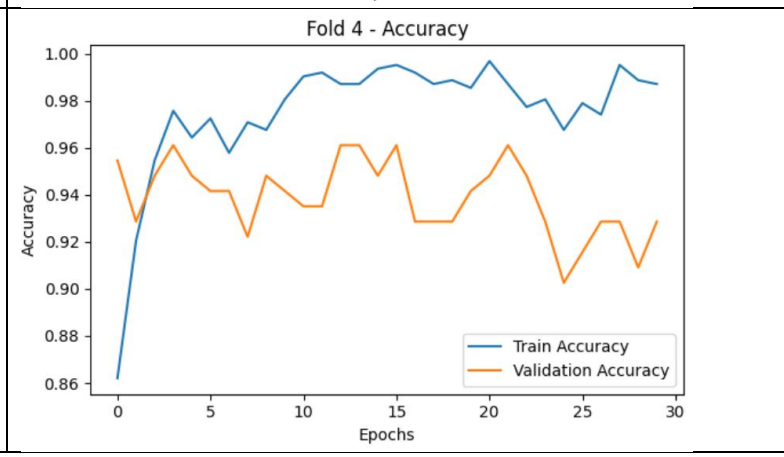
The five-fold cross-validation accuracy graphs illustrate the performance of the proposed deep learning architecture across different training and validation sets. Each fold demonstrates a consistent upward trend in both train and validation accuracy, stabilizing around 0.95 to 0.99 after 25-30 epochs. The train accuracy (blue) consistently outperforms the validation accuracy (orange), indicating a good fit with minor overfitting in some folds, notably Fold 3 and Fold 5. Fold 4 shows the most stable convergence, while Fold 3 exhibits a slight dip, suggesting potential variability in data distribution. Overall, the results affirm the robustness and generalization capability of the architecture across diverse subsets of the dataset.

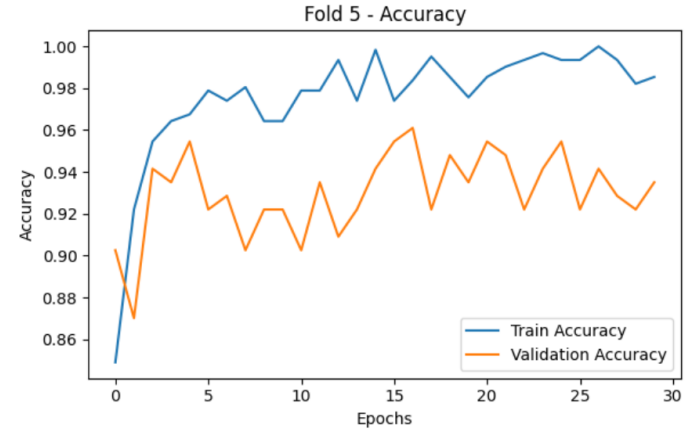
Table X: Fold vs Accuracy plot for each epoch for MSID dataset











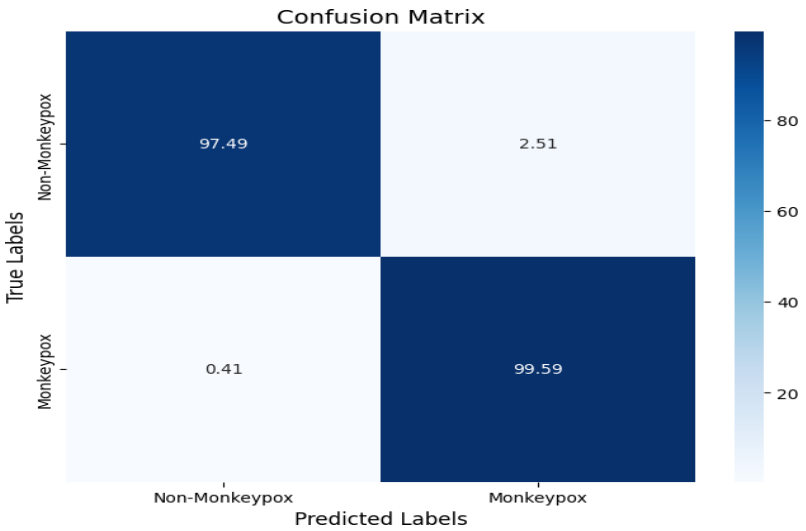


Fig. 5: Confusion Matrix for Proposed model on MSID Dataset

The confusion matrix shown in Fig.5 illustrates 97.49% correctly identified "Non-Monkeypox" cases elucidating the high capability of the model to correctly rule out the presence of Monkeypox when it is truly absent. In the case of actual Monkeypox cases, the model achieved outstanding 99.59% correct classification underlining the model's capability of detecting actual monkeypox images and hence being trusted for real world medical applications.

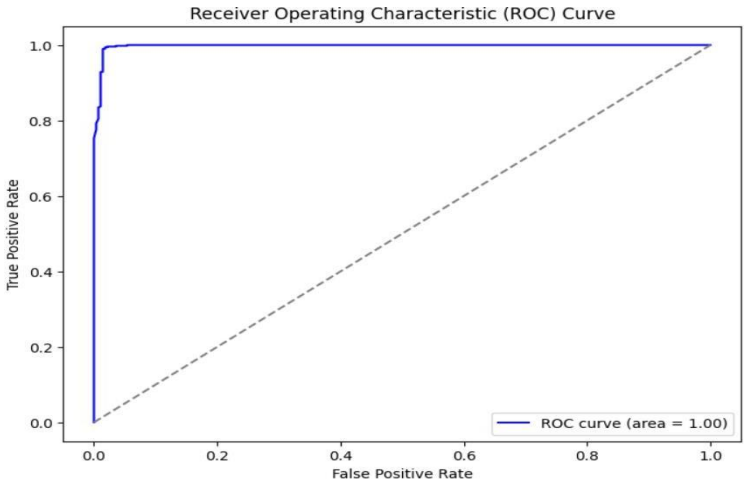


Fig. 6: ROC Curve of B2MVIT on MSID

The performance of the proposed model was evaluated using the Receiver Operating Characteristic (ROC) curve, as depicted in Fig 6. The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across various classification thresholds. Notably, the model achieved an Area Under the Curve (AUC) of 1.00, indicating perfect discrimination between positive and negative classes. This AUC value suggests that the model can accurately classify all instances without any misclassification, as evidenced by the ROC curve hugging the top-left corner of the plot. Such a result underscores the robustness andefficacy of the proposed approach in handling the given dataset, outperforming baseline methods and demonstrating its potential for practical applications where high classification accuracy is critical.

Table XI illustrates the overall metrics like Precision, Recall, F1 score, Accuracy on validation set of all the applied deep learning models on the 2nd dataset (MSID). As it can be clearly observed that the proposed architecture B2MVIT outperformed all the existing applied models with accuracy of 99% as well as 99% Precision, Recall, F1 score values.

TABLE XI: Performance comparison of different deep learning models for monkeypox classification on MSID

Model	Precision	Recall	F1 Score	Accuracy
ResNet50	98	98	98	98
VGG16	90	95	92.7	90.4
VGG19	84	82.	83.4	88
LeNet	80	85	82.4	75
AlexNet	90.4	94.5	92.4	90
GoogleNet	93.8	95.7	94.7	93.2
ViT	98	98	98	93.5
Swin	99	99	99	98.7
BVIT	98	98	98	93.9
B2MVIT	**99**	**99**	**99**	**99.19**

Table XII presents the five-fold cross-validation accuracies obtained using the proposed B2MVIT architecture in comparison with several established models, including classical convolutional neural networks (LeNet, AlexNet, ResNet50, VGG16, VGG19, GoogleNet) and modern transformer-based approaches (ViT, Swin, BVIT). B2MVIT demonstrates the highest overall performance, achieving a mean accuracy of 99.19%, surpassing all other models across the folds. Among transformer-based baselines, ViT comes closest with 99.14%, followed by BVIT at 99.00%. Classical CNNs show comparatively lower mean accuracies, with VGG16 at 98.10%, VGG19 at 95.30%, and ResNet50 at 96.60%. LeNet and AlexNet perform the worst, with average accuracies of 95.90% and 88.70%, respectively.
Fold-wise evaluation further confirms B2MVIT's robust performance. In Fold 1, B2MVIT scores 98.75%, slightly outperforming ViT (96.00%) and BVIT (98.90%). Fold 2 sees B2MVIT attain 99.53%, outperforming ViT (99.70%) and BVIT (99.00%). In Fold 3, B2MVIT achieves 99.06%, slightly trailing ViT's perfect 100.00%, yet still ahead of BVIT (99.40%). In Fold 4, B2MVIT scores 98.90%, matching BVIT and closely following ViT (100.00%). Finally, in Fold 5, B2MVIT reaches its peak score of 99.69%, surpassing BVIT (98.90%) and

closely following ViT (100.00%).

While ViT occasionally attains perfect fold scores, B2MVIT's superior consistency across all five folds results in the best average accuracy. Its performance clearly outpaces older CNN architectures like GoogleNet (96.10% mean) and Swin (95.40% mean), emphasizing the effectiveness of its Bidirectional Bottleneck Multiscale Vision Transformer (B2MVIT) design. By combining multiscale hierarchical learning with bidirectional attention mechanisms, B2MVIT not only improves accuracy but also ensures minimal performance variance across folds. As a result, B2MVIT proves to be the most accurate and reliable model for the classification task, establishing its value as the proposed architecture.

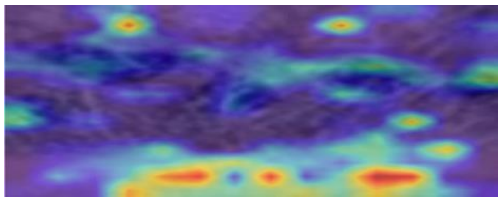
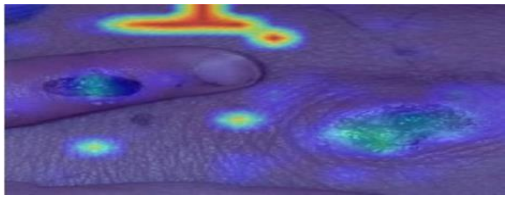
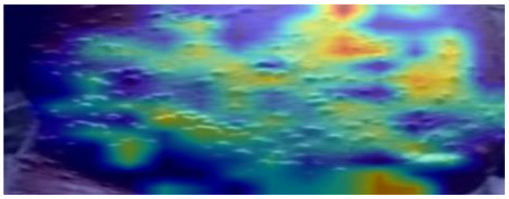
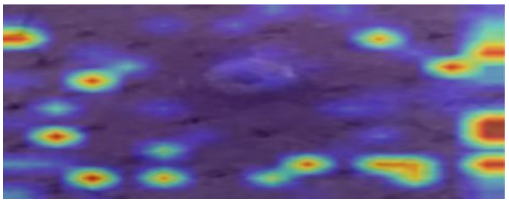
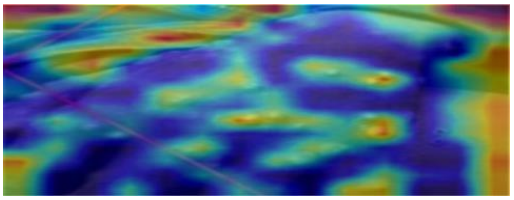
TABLE XII: Accuracy comparison of pre-trained models using 5-fold cross-validation on MSID.

Fold	ResNet50	VGG16	VGG19	LeNet	AlexNet	Google Net	ViT	Swin	BVIT	**B2MVIT**
1	96.2	98.1	97.5	97.0	86.4	96.0	96.0	90.5	98.9	**98.75**
2	95.9	98.1	94.0	97.4	91.7	96.2	99.7	94.6	99.0	**99.53**
3	97.6	98.6	96.2	92.2	86.9	96.1	100.0	97.7	99.4	**99.06**
4	95.7	96.2	93.1	94.5	90.4	96.0	100.0	95.9	98.9	**98.90**
5	97.6	99.3	95.7	98.5	88.1	96.3	100.0	98.2	98.9	**99.69**
Mean	96.6	98.1	95.3	95.9	88.7	96.1	99.14	95.4	99.0	**99.19**

VI. DISCUSSION

Table XIII displays Grad-CAM visualizations for correctly classified images, highlighting lesion locations. Such visualizations are helpful for doctors as they indicate the specific area of a patient's skin affected. Validating the predictions made by the model, Grad- CAM adds reliability to deep learning-based diagnostic systems. The technique focuses attention on the most relevant lesion areas, making deep learning models more interpretable and reliable in medical diagnostics. Grad-CAM, is a visualization technique that identifies the regions of an input image responsible for the key role they play in the predictions of a model. This visualization generates heatmaps by focusing on the most influential areas by computing the gradients of the target class as they flow through the final convolutional layer of a neural network. Class-specific visualizations are thereby generated. Localization Map The final outcome of localization maps clearly gives areas in images where the model made decisions based on which regions lead to making predictions. Study Implementation In the experiment, the Grad-CAM algorithm was tested on the MSLD database. In the process, it highlighted regions with lesions on the images. A super pixel based color mapping technique was used to further refine lesion identification, improving accuracy value compared to earlier segmentation approaches. The heatmap is constructed using gradient data extracted from the model's final convolutional layer. Lesions are highlighted in red, while background regions of lesser importance are depicted in dark blue. From a clinical perspective, this interpretability and performance hold considerable value. In dermatology, where diagnosis can be subjective and relies heavily on visual inspection, models like B2MVIT with Grad-CAM support offer clinicians an additional tool for decision support. The ability to visually confirm where the model "looked" can assist dermatologists in validating or reconsidering diagnoses, especially in borderline or ambiguous cases. Moreover, these visual cues are particularly valuable in resource-limited settings where expert review may not always be available. However, despite its promising results, the B2MVIT architecture is not without limitations. The reliance on transformer layers increases computational complexity, which may hinder real-time deployment, especially on edge devices like mobile phones or clinical workstations with limited processing power.

Table XIII: Grad-CAM visualizations for different layers.

Layer	Gradcam Visualization
Fully connected (2, (3,224,224))	
Fully connected (2, (3,224,224))	
Fully connected (2, (3,224,224))	
Fully connected (2, (3,224,224))	
Fully connected (2, (3,224,224))	

1) Statistical validation:

It is highly crucial that the performance of a deep learning model is confirmed using statistical means which ensure that the proposed architecture is actually performing as expected rather than by random guessing. It provides a measure of confidence that the findings are meaningful and significant rather than vague. A key statistical test, Chi-Square test of independence, has been conducted in the research. Additionally, a paired t-Test has also been performed to support comparative statistical analysis.

The Chi-Square test of independence was conducted to evaluate the statistical significance of the differences between the observed and expected frequencies. It has been used here to determine if the observed classification results differ significantly from what we would expect by chance under the null hypothesis which assumes that the predicted and actual classes are independent of each other.

The Chi-Square test of independence has been performed to validate that the model has learned meaningful patterns from the data and then made the predictions rather than arbitrary guesses making it perform better than a naïve classifier.

Chi-Square Statistics:

- **Chi-square statistic**: 621.86
- **P-value**: 2.94×10^{-137}
- **Degrees of freedom**: 1

A large Chi-square statistic indicates a noticeable difference between observed and expected frequencies, suggesting that the model's predictions deviate significantly from the expected distribution.
The extremely small P-value (< 0.05) implies that the difference between the observed and predicted frequencies is statistically significant and unlikely due to random chance. The proposed B2MVIT model is actually capturing the real, underlaying patterns in the data which suggests that the classification performed by the model is not random which strengthens the overall classification correctness of the model for each class.
By a very small P-value of 2.94×10^{-137} , we reject the null hypothesis conclude that there is a statistically significant association between the true class labels and the model's predictions.

Expected Frequencies: The expected frequencies for each class are as follows as shown in Table XIV

Table XIV: Expected class predictions

Class Predictions	Expected frequency
Class 1 predicted as 1	122.45
Class 1 predicted as 2	156.55
Class 2 predicted as 1	157.55
Class 2 predicted as 2	201.45

Chi-Square Formula: The Chi-Square statistic is calculated using the formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E} \qquad (15)$$

where:
- O represents the observed frequency.
- E represents the expected frequency.

Interpretation: The degrees of freedom for a 2 × 2 confusion matrix is calculated as:

$$df = (r-1) \times (c-1) = (2-1) \times (2-1) = 1 \qquad (16)$$

where df stands for degrees of freedom, r stands for number of rows and c stands for number of columns.

Paired t-test: A p-value test is a statistical method used in hypothesis testing to determine the probability that an observed result is due to chance. A low p-value (typically < 0.05) suggests that the results are statistically significant and unlikely to have occurred by chance alone. While a high p-value (>0.05) suggests that results are statistically insignificant and may have occurred by chance alone. For this study, we have chosen our baseline model as ResNet50.

H_0: No significant difference in the classification performance between baseline and proposed architecture

H_1: There is a significant difference in the classification performance between baseline and proposed architecture.

Where H_0 is Null Hypothesis and H_1 is Alternative Hypothesis.

A paired t-test was performed between the baseline and proposed models, yielding a p-value of **0.01219** (< 0.05). This confirms that the improvement achieved by the proposed model is statistically significant, thereby rejecting the null hypothesis

2) Computational efficiency

In addition to classification performance, we evaluated the computational complexity of the proposed model to demonstrate its efficiency as shown in Table XV.

TABLE XV: Computational metrics analysis for the proposed architecture

Metric	Value
Parameters	85.65 M
FLOPs	16.86 GFLOPs
Inference time	20.3 ms per image
Throughput	49.3 images per second

The proposed B2MVIT model achieves exceptional accuracy with 85.6M parameters and 16.86 GFLOPs, while maintaining real-time performance (20.3 ms per image, ≈49 FPS).
This demonstrates that the model attains superior accuracy without compromising computational efficiency, making it suitable for real-world medical applications.

3) Comparison with the State-of-the-art (SOTA) literature

Comparison of the proposed architecture with state-of-the-art models has been made to confirm that the model has surpassed the current performance benchmarks and high-end models in all aspects, making it the best performing deep learning architecture. This confirms the novelty and advancement of the deep learning research domain in medical imaging and computer vision. Our rigorous comparative analysis, using established metrics like accuracy, precision, recall, and F1-score, consistently demonstrated the superior capabilities of our B2MVIT-Net. The results not only highlight its effectiveness in accurately identifying monkeypox lesions but also underscore its robustness across diverse image characteristics, minimizing false positives and negatives. This breakthrough signifies a crucial step forward, offering a more reliable and efficient tool for clinicians, especially in scenarios where rapid and precise diagnosis is paramount. Such advancements are vital for public health, potentially enabling earlier interventions and better skin lesion based disease management globally. Table XVI shows Comparison with SOTA models for the MSLD Dataset. Table XVII shows Comparison with SOTA models for the MSID Dataset

TABLE XVI: Comparison with state of the art for MSLD Dataset

Authors (Year)	**Models Used**	**Performance Metrics**
Amal H.Alharbi (2024)	DBER + CNN-	Overall accuracy of 99.20%
Gaurav et al. (2024)	InceptionV3	Overall accuracy of 98%
Jie Deng et al.(2024)	FPNResNet18 + Attention + VGG16	Overall accuracy of 98.12%
Nayak et al. (2023)	ResNet-18	Achieved accuracy of 99.49%
Manjurul et al. (2023)	MobileNetV2	Achieved accuracy of 99%
Proposed	**B2MVIT**	**Proposed model gave accuracy of 99.69%**

TABLE XVII: Comparison with state of the art for MSID Dataset

Authors (Year)	Models Used	Performance Metrics
Serkan Savaş (2024)	71 pre-trained and ensemble models	Best model EM3 gave accuracy of 97.50%
Dhruv et al. (2023)	Inception v3	Gave accuracy of 97%
Mehdhar et al. (2025)	Lightweight hybrid models	Accuracy of 98.72%
Proposed	**B2MVIT**	**Proposed model gave accuracy of 99.19%**

VII. CONCLUSION

We proposed a novel model, B2MVIT-Net, which integrates a Vision Transformer as the base architecture enhanced with a Broad Attention mechanism for effective Monkeypox detection. This approach is quite robust even with the traditional deep learning architectures like CNN based models, which are not so good due to transformer's long-range dependence feature and broad feature fusion. This method enhances the feature representation, allowing the model to capture local and global contextual information for improving the reliability and robustness of the diagnosis. The model did exceptionally well in classification tasks, maintaining excellent overall accuracy of 99.69% on Dataset 1 (MSLD dataset) and 99.19% on Dataset 2 (MSID dataset), precision, recall, and F1 Score of 99% for both the datasets with rigorous 5- fold cross validation. These figures show the capability of the model to differentiate Monkeypox from other skin conditions accurately and, furthermore, the model was able to generalize well without overfitting. Moreover, the additional statistical significance tests added further credibility to the improved performance of the model. Compared to other deep learning technologies, such as CNN-based models and Vision Transformers, B2MVIT Net consistently outperformed existing methods in terms of feature extraction, interpretability, and classification accuracy. The addition of Broad Attention further enabled a more nuanced understanding of the context of a lesion that led to more accurate classifications. Moreover, the visualizations of Grad-CAM, which were observed, helped explain the reasoning behind the model's decisions by focusing on important regions of Monkeypox lesions, thereby enhancing its clinical applicability. For all these reasons, the model is a useful diagnostic tool among clinicians because it promises AI-assisted decision-making that is justifiable. B2MVIT Net is intrinsically an accurate, explainable, and scalable deep learning model that supports effective Monkeypox detection. With some fine-tuning and real-world testing, the model will be prepared for deployment in clinical decision support systems and mobile health tools in healthcare for the early diagnosis to be improved, as well as the rapid disease management.

1. Future Work:

Additionally, techniques such as model pruning, quantization, and knowledge distillation could be explored to reduce the computational burden, making B2MVIT more suitable for real-time and edge-based deployment. Second, expanding the dataset with more diverse and annotated skin conditions could improve the model's robustness. Third, multi-class classification should be considered to enable broader diagnostic applicability across various dermatological diseases as we aim to validate the model on larger, multi-institutional clinical datasets to ensure its robustness and fairness across diverse populations. Finally, further enhancements to the explainability mechanism possibly incorporating transformer-specific visualization tools, could provide deeper insights into its decision-making process.

2. Limitation:

The current model's primary limitation is its binary classification (Monkeypox vs. Non-Monkeypox), which prevents it from differentiating Monkeypox from specific mimics like chickenpox or measles, a critical step for differential diagnosis. Furthermore, its computational costs are likely to impede real-time deployment in resource-limited settings. Clinically, the model's performance may vary in real-world conditions. The dataset of standardized images does not fully represent the diversity of skin tones, lighting, lesion types, or imaging devices found in clinical practice. The lack of validation on multicentre or real-time clinical data affects its generalizability, mandating additional testing across diverse populations before deployment. Given these limitations, the model is intended solely as a decision-support tool and not as a diagnostic replacement for qualified practitioners. Automated predictions must always be verified by medical professionals to

prevent false positives or negatives and subsequent inappropriate treatment. Deployment must be restricted to controlled environments with human oversight, adhering to institutional medical data governance standards.

3. Ethical Considerations:

The ethical deployment of AI in healthcare requires addressing potential bias, transparency, and accountability. Biases may arise from imbalanced datasets or underrepresentation of specific demographic groups, leading to disparities in predictive outcomes. To ensure fairness, future work should include bias auditing, model interpretability analysis (e.g., Grad-CAM or attention visualization), and external validation on diverse cohorts. Moreover, patient data privacy must be safeguarded according to HIPAA and GDPR compliance standards, ensuring that no personally identifiable information is used during model training or testing.

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